COURSE 2 · SECTION 7 · EXERCISE

# Two dates, one bridge

Section 7 (videos S07.1 and S07.2) · Data model + DAX for engineering data

**Start:** Section08_1_Start.pbix or your Engxtech_S07_2.pbix; save a copy as Engxtech_S07_Exercise.pbix

**Time:** about 25 min

**Data:** the course data folder

**DESIGN QUESTION**

How many changes were requested and how many were closed in each year, and how many changes touch the P300 seat ring: counted by primary part, and counted by every part a change affects?

**Extra practice.** This exercise is extra practice: a longer task than the video's short "Your turn" questions, which are answered in the video itself. Do it after the video, in a copy of your file.

Changes has two dates that matter: RequestDate and CloseDate. The Dates table from the video has one active relationship to RequestDate and one inactive to CloseDate. ChangeAffectedItems is the bridge between Changes and Parts; the PrimaryPart line to Parts is inactive.

**New in this exercise** (not shown in the video; one line each):

- **VALUES ( column )**: returns the distinct values of the column that are left after the current filters.
- **TREATAS ( table, column )**: applies the values of the table as a filter on another column, as if a relationship existed (stretch task only).

## Your tasks

1. In your exercise copy, check Model view: Dates[Date] 1:* Changes[RequestDate] (active), Dates[Date] 1:* Changes[CloseDate] (inactive), Changes 1:* ChangeAffectedItems, Parts 1:* ChangeAffectedItems, and Changes[PrimaryPart] *:1 Parts[PartID] (inactive).
2. Matrix 1: rows Dates[Year]; values **Changes Requested** and **Changes Closed** (both from the video, in _Measures).
3. Matrix 2: rows Parts[PartID]; values **Changes Affecting Part** and **Changes by Primary Part**. Read P300.
4. Stretch (optional): new measure in _Measures: **Primary Part Changes (TREATAS)**, the primary-part count without the inactive relationship. It must match Changes by Primary Part on every row.
5. Read the Closed % card and the ChangeAffectedItems row count.

## Check your answer

| Check | Expected |
|---|---|
| Change Count, all | **178** |
| Requested in 2025 / 2026 | **81 / 97** |
| Closed in 2025 / 2026 (USERELATIONSHIP) | **66 / 88** |
| Closed % | **86.5 %** |
| P300: Changes Affecting Part (bridge) | **45** |
| P300: Changes by Primary Part | **26** |
| ChangeAffectedItems rows | **304** |

**Hint.** TREATAS takes the PartIDs that are in the filter and applies them to Changes[PrimaryPart] as if a relationship existed. If every part shows 178, the filter from Parts never reached Changes.

Stuck? The worked solution with the full DAX and M is in Solutions/S07_Solution.pdf.